

Frequency Response Part 2

EGR 345

GVSU

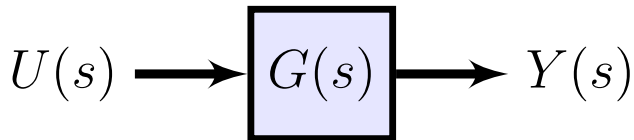
It was the best of lectures, it was the worst of lectures....

- I love talking about frequency response and Bode
- I need to working on breaking my lectures into chunks
 - more examples
 - planning on where to break things between lectures

Key Points from Last Lecture

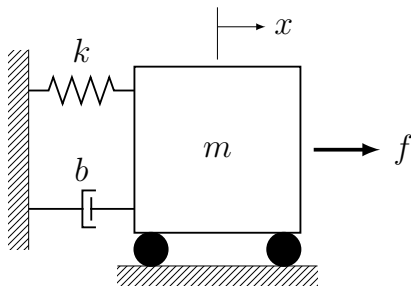
- The transfer function does not change based on the input
 - yes, we solve for $G(s) = Out(s)/In(s)$, but we assume $In(s)$ is unknown
- frequency response (Bode) only applies to stable systems
 - transients die out

What do we mean by frequency response?



- $u(t) = \sin(\omega t)$
- $y_{ss}(t) = M \sin(\omega t + \phi)$
- assuming $G(s)$ is stable
 - transient terms die out

Applying this to SDOF MSD



$$G(s) = \frac{1/m}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad \text{and} \quad U(s) = \frac{\omega}{s^2 + \omega^2}$$

$$Y(s) = \frac{1/m}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot \frac{\omega}{s^2 + \omega^2}$$

PFE

$$Y(s) = \frac{1/m}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot \frac{\omega}{s^2 + \omega^2} = \frac{C_1\omega + C_2s}{s^2 + \omega^2} + \frac{C_3\omega_d + C_4(s+a)}{(s+a)^2 + \omega_d^2}$$